

To check the Docker version, use the following command:

```
[root@localhost ~] # docker version
```

How to create a LAMP stack with Docker

We are going to create a LAMP stack on a CentOS VM. However, you can work on different variants as well. First, let's get the latest CentOS image. The command below will help us to do so:

```
[root@localhost ~] # docker pull centos:latest
```

Next, let's make sure that we can see the image by running the following code:

```
[root@localhost ~] # docker image centos
REPOSITORY TAG IMAGE ID CREATED VIRTUAL SIZE
centos latest 0c752394b855 13 days ago 124.1 MB
```

Running a simple bash shell to test the image also helps you to start a new container:

```
[root@localhost ~] # docker run -i -t centos /bin/bash
```

If everything is working properly, you'll get a simple bash prompt. Now, as this is just a base image, we need to install the PHP, MySQL and the LAMP stack:

```
[root@localhost ~] # yum install php php-mysql mysql-server httpd
```

The container now has the LAMP stack. Type 'exit' to quit from the bash shell.

We are going to create this as a golden image, so that the next time we need another LAMP container, we don't need to install it again.

Run the following command and please note the 'CONTAINER ID' of the image. In my case, the ID was '4de5614dd69c':

```
[root@localhost ~] # docker ps -a
```

The ID shown in the listing is used to identify the container you are using, and you can use this ID to tell Docker to create an image.

Run the command below to make an image of the previously created LAMP container. The syntax is `docker commit <CONTAINER ID> <name>`. I have used the previous container ID, which we got in the earlier step:

```
[root@localhost ~] # docker commit 4de5614dd69c lamp-image
```

Run the following command to see your new image in the list. You will find the newly created image 'lamp-image' is shown in the output:

```
[root@localhost ~] # docker images
REPOSITORY TAG IMAGE ID CREATED VIRTUAL SIZE
lamp-image latest b71507766b2d 2 minutes ago 339.7 MB
centos latest 0c752394b855 13 days ago 124.1 MB
```

Let's log in to this image/container to check the PHP version:

```
[root@localhost ~] # docker run -i -t lamp-image /bin/bash
bash-4.1# php -v
PHP 5.3.3 (cli) (built: Dec 11 2013 03:29:57)
Zend Engine v2.3.0 Copyright (c) 1998-2010 Zend Technologies
```

Now, let us configure Apache.

Log in to the container and create a file called `index.html`.

If you don't want to install VI or VIM, use the `Echo` command to redirect the following content to the `index.html` file:

```
<?php echo "Hello world"; ?>
```

Start the Apache process with the following command:

```
[root@localhost ~] # /etc/init.d/httpd start
```

And then test it with the help of `browser/curl/links` utilities.

If you're running Docker inside a VM, you'll need to forward port 80 on the VM to another port on the VM's host machine. The following command might help you to configure port forwarding. Docker has the feature to forward ports from containers to the host.

```
[root@localhost ~] # docker run -i -t -p :80 lamp-image /bin/bash
```

For detailed information on Docker and other technologies related to container virtualisation, check out the links given under 'References'. 

References

- [1] Docker: <https://docs.docker.com/>
- [2] LXC: <https://linuxcontainers.org/>

By: Pradyumna Dash

The author is an independent consultant, and works as a cloud/ DevOps architect. An open source enthusiast, he loves to cook good food and brew ideas. He is also the co-founder of the site <http://www.sillyoan.org/>